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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 12

**PHYSICAL SCIENCES: CHEMISTRY (P2)
FISIESE WETENSKAPPE: CHEMIE (V2)**

NOVEMBER 2021

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 21 pages.
*Hierdie nasienriglyne bestaan uit 21 bladsye.***

QUESTION 1/VRAAG 1

- 1.1 D ✓✓ (2)
- 1.2 D ✓✓ (2)
- 1.3 A ✓✓ (2)
- 1.4 B ✓✓ (2)
- 1.5 D ✓✓ (2)
- 1.6 D ✓✓ (2)
- 1.7 C ✓✓ (2)
- 1.8 B ✓✓ (2)
- 1.9 A ✓✓ (2)
- 1.10 B ✓✓ (2)
- [20]**

QUESTION 2/VRAAG 2

- 2.1 A compound that contains a double bond/multiple bond/does NOT contain only single bonds (between C atoms). ✓✓ **(2 or 0)**
'n Verbinding wat dubbelbindings/meervoudige bindings/NIE net enkelbindings (tussen C-atome) bevat NIE. (2 of 0) (2)
- 2.2
- 2.2.1 B / E ✓ (1)
- 2.2.2 Carbonyl (group bonded to two C atoms) ✓
Karboniel(groep gebind aan twee C-atome) **ACCEPT/AANVAAR**
Ketone/Ketoon (1)
- 2.2.3 F ✓✓ (2)
- 2.2.4 2,5-dichloro-3-methylhexane/2,5-dichloro-3-metielheksaan

Marking criteria:

- Correct stem i.e. hexane. ✓
- All substituents (dichloro and methyl) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- Korrekte stam d.i. heksaan. ✓
- Alle substituenten (dichloro en metiel) korrek geïdentifiseer. ✓
- IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas. ✓

(3)

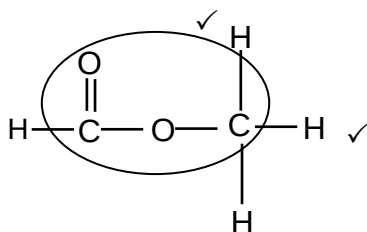
2.2.5 C_nH_{2n} ✓ (1)

2.3 Compounds with the same molecular formula, ✓ but different functional groups/homologous series. ✓
Verbindings met dieselfde molekulêre formule, maar verskillende funksionele groepe/homoloë reekse. (2)

2.4

2.4.1 Carboxylic acids/Karboksielsure ✓ (1)

2.4.2



Marking criteria/Nasienkriteria:

- Whole structure correct/
Hele struktuur korrek: $\frac{2}{2}$
- Only functional group correct/Slegs
funksionele groep korrek: Max/Maks.: $\frac{1}{2}$

IFIINDIEN

- More than one functional group:
Meer as een funksionele groep: $\frac{0}{2}$

IFIINDIEN

- Molecular formula/Molekulêre formule $\frac{0}{2}$
- Condensed structural formula /Gekondenseerde struktuurformule $\frac{1}{2}$

2.5

2.5.1 Ethanol/Etanol ✓ (1)

2.5.2 E ✓ **ACCEPT/AANVAAR:** C_2H_4 (1)

2.5.3 (Concentrated) sulphuric acid/ H_2SO_4 /(concentrated) phosphoric acid/ H_3PO_4 ✓
 (Gekonsentreerde) swawelsuur/ H_2SO_4 /(gekonsentreerde) fosforsuur/ H_3PO_4 (1)
[18]

QUESTION 3/VRAAG 3

3.1

Marking criteria/Nasienkriteria:

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. *Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.*

The temperature at which solid and liquid phases are in equilibrium. ✓✓
Die temperatuur waarby die vastestof- en vloeistoffases van 'n stof in ewewig is.

(2)

3.2

Marking criteria

- Identification of independent variable. ✓
- Stating the relationship between dependent and independent variable. ✓

Nasienkriteria

- *Identifikasie van onafhanklike veranderlike.* ✓
- *Stel verwantskap tussen afhanklike en onafhanklike veranderlikes.* ✓
- As the chain length/number of C atoms/molecular mass/surface area/strength of the intermolecular forces ✓ increases, the melting points increase. ✓
OR
- As the chain length/ number of C atoms/molecular mass/surface area/strength of the intermolecular forces ✓ decreases, the melting points decrease. ✓
- *Wanneer die kettinglengte/aantal C-atome/molekulêre massa/oppervlak-area/sterkte van intermolekulêre kragte ✓ toeneem, neem die smeltpunte toe.*
OF
- *Wanneer die kettinglengte/aantal C-atome/molekulêre massa/oppervlak-area/sterkte van intermolekulêre kragte afneem, neem die smeltpunte af.*

(2)

3.3

London forces ✓
Londonkragte

ACCEPT/AANVAAR

Dispersion forces/induced dipole forces
Dispersiekragte/geïnduseerde dipoolkragte

(1)

3.4

3.4.1

Liquid/Vloeistof ✓

(1)

3.4.2

Solid/Vaste stof ✓

(1)

3.5

3.5.1

Equal to/Gelyk aan ✓

Same molecular formula/Isomers/same number and types of atoms/same number of C and H atoms ✓

Dieselfde molekulêre formule/Isomere/dieselfde aantal en soort atome/dieselfde aantal C- en H-atome

(2)

3.5.2

Lower than/Laer as ✓

(1)

3.5.3

Marking criteria:

- Compare structures. ✓
- Compare the strength of intermolecular forces. ✓
- Compare the energy required to overcome intermolecular forces. ✓

2,2-dimethylbutane:

- **Structure:**
More branched/more compact/more spherical/smaller surface area (over which intermolecular forces act). ✓
- **Intermolecular forces:**
Weaker/less intermolecular forces/Van der Waals forces/London forces/dispersion forces. ✓
- **Energy:**
Lesser energy needed to overcome or break intermolecular forces/Van der Waals forces. ✓

OR

Hexane

- **Structure:**
Longer chain length/unbranched/less compact/less spherical/larger surface area (over which intermolecular forces act). ✓
- **Intermolecular forces:**
Stronger/more intermolecular forces/Van der Waals forces/London forces/dispersion forces. ✓
- **Energy:**
More energy needed to overcome or break intermolecular forces/Van der Waals forces. ✓

Nasienkriteria:

- *Vergelyk strukture* ✓
- *Vergelyk die sterkte van intermolekulêre kragte.* ✓
- *Vergelyk die energie benodig om intermolekulêre kragte te oorkom.* ✓

2,2-dimetielbutaan:

- **Struktuur:**
Meer vertak/meer kompak/meer sferies/kleiner oppervlak (waaroor intermolekulêre kragte werk). ✓
- **Intermolekulêre kragte:**
Swakker/minder intermolekulêre kragte/Van der Waalskragte/London-kragte/dispersiekragte. ✓
- **Energie:**
Minder energie benodig om intermolekulêre kragte/Van der Waalskragte/dispersiekragte/Londonkragte te oorkom/breek. ✓

OF

Heksaan

- **Struktuur:**
Langer kettinglengte/onvertak/minder kompak/minder sferies/groter oppervlak (waaroor intermolekulêre kragte werk). ✓
- **Intermolekulêre kragte:**
Sterker/meer intermolekulêre kragte/Van der Waalskragte/Londonkragte/dispersiekragte. ✓
- **Energie:**
Meer energie benodig om intermolekulêre kragte/Van der Waalskragte/dispersiekragte/Londonkragte te oorkom/breek. ✓

(3)
[13]

QUESTION 4/VRAAG 4

4.1

4.1.1 Substitution/Hydrolysis ✓
Substitusie/Hidrolise

(1)

4.1.2 Primary (alcohol) ✓

ANY ONE:

- The C atom of the functional group is the terminal C atom.
- The C-atom bonded to the hydroxyl/-OH is bonded to (only) one other C-atom. ✓
- The hydroxyl/-OH is bonded to a C-atom which is bonded to two hydrogen atoms.
- The hydroxyl/-OH is bonded to a primary C atom/terminal C atom/first C atom.

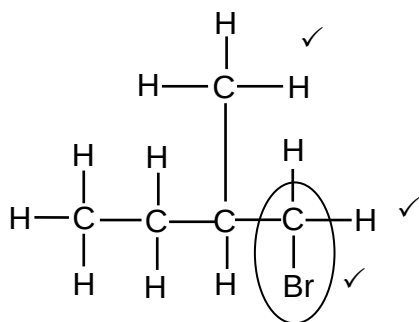
Primêre (alkohol) ✓

ENIGE EEN:

- Die C-atoom van die funksionele groep is die terminale C-atoom.
- Die C-atoom wat aan die hidroksiell/-OH gebind is, is aan (slegs) een ander C-atoom gebind. ✓
- Die hidroksiell/-OH is gebind aan 'n C-atoom wat aan twee waterstofatome gebind is.
- Die hidroksiell/-OH is aan 'n primêre C-atoom/terminale C-atoom/eerste C-atoom gebind.

(2)

4.1.3

**Marking criteria:**

- Four C atoms in longest chain. ✓
- One methyl substituent on C2. ✓
- Bromo substituent on C1. ✓

Nasienkriteria:

- Vier C-atome in langste ketting. ✓
- Een metielsubstituent op C2. ✓
- Broomsustituent op C1. ✓

IF/INDIEN

Any error e.g. omission of H atoms, condensed or semi structural formula/Enige fout bv. weglating van H-atome, gekondenseerde of semi-struktuurformule. Max/Maks.: $\frac{2}{3}$

(3)

4.1.4 Elimination/dehydrohalogenation/dehydrobromination ✓
Eliminasie/dehidrohalogenering/dehidrohalogenasie/dehidrobrominasie/
dehidrobromonering

(1)

4.1.5 Alkenes/Alkene ✓

(1)

4.1.6 Addition/Addisie ✓

(1)

4.1.7 2-bromo-2-methylbutane ✓
2-bromo-2-metielbutaan ✓

(2)

4.2

NOTE/LET WEL:

- Penalise only once for the use of structural formulae or molecular formulae.
- *Penaliseer slegs een keer vir die gebruik van struktuurformules of molekulêre formules.*

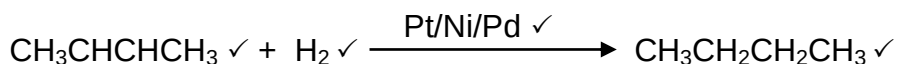
4.2.1

Marking criteria:

- Correct condensed structure for but-2-ene. ✓
- React but-2-ene with $H_2/H - H$. ✓
- Indicate the catalyst Pt/Ni/Pd on arrow/at the equation. ✓
- Correct condensed formula for butane as product. ✓

IF: Any additional products or reactants - minus 1 mark**Nasienkriteria:**

- *Korrekte gekondenseerde struktuur vir but-2-ene*. ✓
- *Reageer but-2-ene met $H_2/H - H$* . ✓
- *Dui die katalisator Pt/Ni/Pd op die pyl/by die vergelyking aan*. ✓
- *Korrekte gekondenseerde formule vir butaan as produk*. ✓

INDIEN: Enige addisionele reaktanse of produkte – minus 1 punt**ACCEPT/AANVAAR**As reactant/reaktans: $CH_3(CH)_2CH_3$ / $CH_3CH = CHCH_3$ / $CH_3 - CH = CH - CH_3$ As product/produk: $CH_3(CH_2)_2CH_3$ / $CH_3 - CH_2 - CH_2 - CH_3$ / $CH_3 - (CH_2)_2 - CH_3$

(4)

4.2.2

Elimination/Cracking ✓

Eliminasie/Kraking

(1)

4.2.3

Propene/1-propene/prop-1-ene ✓✓

Propeen/1-propeen/prop-1-ene

(2)

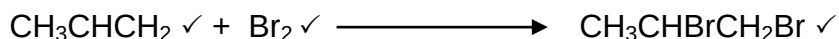
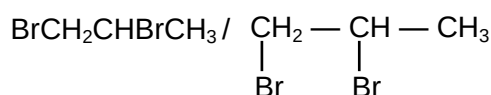
4.2.4

Marking criteria:

- Correct condensed formula for propene as reactant. ✓
- React (propene) with $Br_2/Br - Br$. ✓
- Correct condensed formula for 1,2-dibromopropane as product. ✓

IF: Any additional products or reactants - minus 1 mark**Nasienkriteria:**

- *Korrekte gekondenseerde formule vir propen as reaktans*. ✓
- *Reageer (propen) met $Br_2/Br - Br$* . ✓
- *Korrekte gekondenseerde formule vir 1,2-dibromopropaan as produk*. ✓

INDIEN: Enige addisionele reaktanse of produkte – minus 1 punt**ACCEPT/AANVAAR:**As reactant/reaktans: $CH_3CH = CH_2$ / $CH_2 = CHCH_3$ As product/produk: $CH_3CHBrCH_2Br$ / $CH_3 - \underset{\substack{| \\ Br}}{CH} - \underset{\substack{| \\ Br}}{CH_2}$ /

(3)

[21]

QUESTION 5/VRAAG 5

5.1

NOTE/LET WELGive the mark for per unit time only if in context of reaction rate.*Gee die punt vir per eenheidtyd slegs indien in konteks van reaksietempo.***ANY ONE**

- Change in concentration ✓ of products/reactants per (unit) time. ✓
- Change in amount/number of moles/volume/mass of products or reactants per (unit) time.
- Amount/number of moles/volume/mass of products formed/reactants used per (unit) time.
- Rate of change in concentration/amount of moles/number of moles/volume/mass. ✓✓ **(2 or 0)**

ENIGE EEN

- Verandering in konsentrasie ✓ van produkte/reaktanses per (eenheid) tyd. ✓
- Verandering in hoeveelheid/getal mol/volume/massa van produkte of reaktanses per (eenheid) tyd.
- Hoeveelheid/getal mol/volume/massa van produkte gevorm/reaktanses gebruik per (eenheid) tyd.
- Tempo van verandering in konsentrasie/ hoeveelheid mol/getal mol/volume/ massa. ✓✓ **(2 of 0)**

(2)

5.2

Reaction rate decreases./Concentration of HCl decreases./Concentration of reactant decreases./Reactants are used up/Mass of CaCO₃ decreases or is used up. ✓

Reaksietempo neem af./Konsentrasie van HCl neem af./Konsentrasie van reaktans neem af./Reaktanses word opgebruik./Massa van CaCO₃ neem af of word opgebruik. ✓

(1)

5.3

5.3.1

Exothermic/Eksotermies ✓

(1)

5.3.2

- Gradient increases/becomes steeper. / Curve becomes steeper. ✓
- Reaction rate increases/More (or larger volume) of CO₂ is produced per unit time. ✓
- Temperature increases./Energy is released/Average kinetic energy of the molecules increases. ✓
- *Gradiënt neem toe/word steiler. / Kurwe word steiler. ✓*
- *Reaksietempo neem toe./Meer (of groter volume) CO₂ word produseer per eenheidtyd. ✓*
- *Temperatuur neem toe./Energie word vrygestel./Gemiddelde kinetiese energie van molekule neem toe. ✓*

(3)

5.4

Marking criteria

- $m(\text{pure CaCO}_3) = \frac{82,5}{100} \times 15 \checkmark / V(\text{CO}_2) = \frac{82,5}{100} \times V(\text{CO}_2) \text{ from/uit } 15 \text{ g CaCO}_3$
- Divide by 100 $\text{g} \cdot \text{mol}^{-1}$. $\checkmark$
- Use mol ratio: $n(\text{CO}_2) = n(\text{CaCO}_3)$. $\checkmark$
- Multiply $n(\text{CO}_2)$ by 24 000 $\text{cm}^3/24 \text{ dm}^3$. $\checkmark$
- Final answer: 2 976 cm^3 $\checkmark$
- **Range:** 2880 to 2970 cm^3 / 2,88 to 2,97 dm^3

Nasienkriteria

- $m(\text{suiwer CaCO}_3) = \frac{82,5}{100} \times 15 \checkmark / V(\text{CO}_2) = \frac{82,5}{100} \times V(\text{CO}_2) \text{ uit } 15 \text{ g CaCO}_3$
- Deel deur 100 $\text{g} \cdot \text{mol}^{-1}$. $\checkmark$
- Gebruik molverhouding: $n(\text{CO}_2) = n(\text{CaCO}_3)$. $\checkmark$
- Vermenigvuldig $n(\text{CO}_2)$ met 24 000 $\text{cm}^3/24 \text{ dm}^3$. $\checkmark$
- Finale antwoord: 2 976 cm^3 $\checkmark$
- **Gebied:** 2880 tot 2970 cm^3 / 2,88 tot 2,97 dm^3

OPTION 1/OPSIE 1

$$\begin{aligned}
 m(\text{pure/suiwer CaCO}_3) &= \frac{82,5}{100} \times 15 \checkmark \\
 &= 12,375 \text{ g} \\
 n(\text{pure/suiwer CaCO}_3) &= \frac{m}{M} \\
 &= \frac{12,375}{100} \checkmark \\
 &= 0,124 \text{ mol} \\
 n(\text{CO}_2) &= n(\text{CaCO}_3) \\
 &= 0,124 \text{ mol} \checkmark \\
 V(\text{CO}_2) &= 0,124 \times 24\,000 \checkmark \\
 &= 2\,976 \text{ cm}^3 \checkmark
 \end{aligned}$$

OR/OF

$$\begin{aligned}
 V(\text{CO}_2) &= 0,124 \times 24 \checkmark \\
 &= 2,98 \text{ dm}^3 \checkmark
 \end{aligned}$$

OPTION 2/OPSIE 2IF 15 g PURE CaCO_3 reacts:INDIEN 15 g SUIWER CaCO_3 reageer:

$$\begin{aligned}
 n(\text{CaCO}_3) &= \frac{m}{M} \\
 &= \frac{15}{100} \checkmark \\
 &= 0,15 \text{ mol} \\
 n(\text{CO}_2) &= n(\text{CaCO}_3) \checkmark \\
 &= 0,15 \text{ mol} \\
 n(\text{CO}_2) &= \frac{V}{V_M} \\
 0,15 &= \frac{V}{24\,000} \checkmark / 0,15 = \frac{V}{24} \\
 V &= 3\,600 \text{ cm}^3 / V = 3,6 \text{ dm}^3
 \end{aligned}$$

Actual CO_2 formed:Werklike CO_2 gevorm:

$$\begin{aligned}
 V(\text{CO}_2) &= \frac{82,5}{100} \times 3\,600 / 3,6 \checkmark \\
 &= 2\,976 \text{ cm}^3 / 2,976 \text{ dm}^3 \checkmark
 \end{aligned}$$

(5)

OPTION 3/OPSIE 3IF 15 g PURE CaCO_3 reacts:/INDIEN 15 g SUIWER CaCO_3 reageer:

$$n(\text{CaCO}_3) = \frac{m}{M}$$

$$= \frac{15}{100} \checkmark$$

$$= 0,15 \text{ mol}$$

$$n(\text{CO}_2) = n(\text{CaCO}_3) \checkmark$$

$$= 0,15 \text{ mol}$$

$$n(\text{CO}_2) = \frac{m}{M}$$

$$m(\text{CO}_2) = 0,15 \times 44$$

$$= 6,6 \text{ g}$$

$$82,5 = \frac{m_{\text{actual/werklik}}}{6,6} \times 100 \checkmark$$

$$m_{\text{(actual/werklik)}} = 5,445 \text{ g}$$

$$n(\text{CO}_2) = \frac{m}{M}$$

$$= \frac{5,445}{44}$$

$$= 0,12375 \text{ mol}$$

$$n(\text{CO}_2) = \frac{V}{V_M}$$

$$0,12375 = \frac{V}{24\,000} \checkmark / 0,12375 = \frac{V}{24}$$

$$V = 2\,976 \text{ cm}^3 / 2,976 \text{ dm}^3 \checkmark$$

(5)

5.5 Increases/Toeneem $\checkmark$

(1)

5.6

- More (CaCO_3) particles with correct orientation/exposed./ Greater (exposed) surface area. $\checkmark$
- More effective collisions per unit time./Higher frequency of effective collisions. $\checkmark$
- Meer (CaCO_3)-deeltjies met korrekte oriëntasie/blootgestel./ Groter (blootgestelde) reaksieoppervlakte. $\checkmark$
- Meer effektiewe botsings per eenheid tyd./Hoër frekwensie van effektiewe botsings. $\checkmark$

NOTE/LET WEL

- If explanation in terms of CONCENTRATION: No mark for bullet 1.
Indien verduideliking in terme van KONSENTRASIE: Geen punt vir kolpunt 1.
- Bullets are marked independently./Kolpunte word onafhanklik nagesien.

(2)

[15]

QUESTION 6/VRAAG 6

- 6.1 (The stage in a chemical reaction when the) rate of forward reaction equals the rate of reverse reaction. ✓✓ **(2 or 0)**

OR

(The stage in a chemical reaction when the) concentrations of reactants and products remain constant. **(2 or 0)**

*(Die stadium in 'n chemiese reaksie wanneer die) tempo van die voorwaartse reaksie gelyk is aan die tempo van die terugwaartse reaksie. **(2 of 0)***

OF

*(Die stadium in 'n chemiese reaksie wanneer die) konsentrasies van reaktanse en produkte konstant bly. **(2 of 0)***

(2)

6.2

- 6.2.1 Negative/Negatief ✓

(1)

- 6.2.2 • Increase in temperature favours an endothermic reaction.

Accept: Decrease in temperature favours an exothermic. ✓

- Reverse reaction is favoured./Concentration of reactants increases./Concentration of products decreases. ✓

- (Forward) reaction is exothermic.

Accept: Reverse reaction is endothermic. ✓

- *Toename in temperatuur bevoordeel 'n endotermiese reaksie. ✓*

Aanvaar: *Afname in temperatuur bevoordeel die eksotermiese reaksie.*

- *Terugwaartse reaksie word bevoordeel./Konsentrasie van reaktanse neem toe./Konsentrasie van produkte neem af. ✓*

- *(Voorwaartse) reaksie is eksotermies.*

Aanvaar: *Terugwaartse reaksie is endotermies. ✓*

(3)

6.2.3

CALCULATIONS USING NUMBER OF MOLES**BEREKENINGE WAT GETAL MOL GEBRUIK****Marking criteria:**

- Initial $n(P)$ and $n(Q_2)$ and $n(PQ)$ from table. ✓
- Change in $n(P)$ = equilibrium $n(P)$ – initial $n(P)$. ✓
- USING** ratio: $P : Q_2 : PQ = 2 : 1 : 2$ ✓
- Equilibrium $n(Q_2)$ = initial $n(Q_2)$ + change in $n(Q_2)$ } ✓
Equilibrium $n(PQ)$ = initial $n(PQ)$ - change in $n(PQ)$ }
- Divide **equilibrium** amounts of P and Q_2 and PQ by 2 dm^3 . ✓
- Correct K_c expression (formulae in square brackets). ✓
- Substitution of equilibrium concentrations into K_c expression. ✓
- Final answer: 10,889 ✓

Nasienkriteria:

- Aanvanklike $n(P)$ en $n(Q_2)$ en $n(PQ)$ uit tabel. ✓
- Verandering in $n(P)$ = ewewigs $n(P)$ – aanvanklike $n(P)$. ✓
- GEBRUIK** verhouding: $P : Q_2 : PQ = 2 : 1 : 2$ ✓
- Ewewig $n(Q_2)$ = aanvanklike $n(Q_2)$ + verandering in $n(Q_2)$ } ✓
Ewewig $n(PQ)$ = aanvanklike $n(PQ)$ - verandering in $n(PQ)$ }
- Deel **ewewigshoeveelhede** van P en Q_2 en PQ deur 2 dm^3 . ✓
- Korrekte K_c -uitdrukking (formules in vierkanthakies). ✓
- Vervanging van ewewigskonsentrasies in K_c -uitdrukking. ✓
- Finale antwoord: 10,89 / 10,889 ✓

(3)

OPTION 1/OPSIE 1

	P	Q_2	PQ	
Initial quantity (mol) Aanvangshoeveelheid (mol)	0,8	0,8	3,2	✓(a)
Change (mol) Verandering (mol)	0,4 ✓(b)	0,2	0,4	✓(c)
Quantity at equilibrium (mol)/ Hoeveelheid by ewewig (mol)	1,2	1,0	2,8	✓(d)
Equilibrium concentration ($\text{mol} \cdot \text{dm}^{-3}$) Ewewigskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)	0,6	0,5	1,4	✓(e)

$$K_c = \frac{[PQ]^2}{[Q_2][P]^2} \quad \checkmark (f)$$

$$= \frac{1,4^2}{(0,5)(0,6)^2} \quad \checkmark (g)$$

$$= 10,89 \quad \checkmark (h)$$

No K_c expression, correct substitution/Geen K_c -uitdrukking, korrekte substitusie: Max./Maks. $\frac{7}{8}$

Wrong K_c expression/Verkeerde K_c -uitdrukking:
Max./Maks. $\frac{5}{8}$

OPTION 2/OPSIE 2

	PQ	P	Q ₂	
Initial quantity (mol) <i>Aanvangshoeveelheid (mol)</i>	3,2	0,8	0,8	✓(a)
Change (mol) <i>Verandering (mol)</i>	0,4	0,4 ✓(b)	0,2 ✓(c)	
Quantity at equilibrium (mol)/ <i>Hoeveelheid by ewewig (mol)</i>	2,8	1,2 ✓(d)	1,0	
Equilibrium concentration (mol·dm ⁻³) <i>Ewewigskonsentrasie (mol·dm⁻³)</i>	1,4	0,6	0,5	✓(e)

Reverse reaction

Terugwaartse reaksie:

$$K_c = \frac{[P]^2[Q_2]}{[PQ]^2} \quad \checkmark(f)$$

$$= \frac{(0,6)^2(0,5)}{(1,4)^2} \quad \checkmark(g)$$

$$K_c = 0,09$$

Forward reaction/*Voorwaartse reaksie:*

$$K_c = \frac{1}{0,09} \\ = 10,89 \quad \checkmark(h)$$

No K_c expression, correct substitution/Geen K_c-uitdrukking, korrekte substitusie: Max./Maks. $\frac{7}{8}$

Wrong K_c expression/Verkeerde K_c-uitdrukking: Max./Maks. $\frac{5}{8}$

CALCULATIONS USING NUMBER OF MOLES BEREKENINGE WAT GETAL MOL GEBRUIK

Marking criteria:

- Initial $n(P) = 4 \text{ mol}$ and $n(Q_2) = 2,4 \text{ mol}$ and $n(PQ) = 0$ ✓
- Change in $n(P) = \text{equilibrium } n(P) - \text{initial } n(P) = 2,8 \text{ mol}$. ✓
- USING ratio: $P : Q_2 : PQ = 2 : 1 : 2$ ✓
- Equilibrium $n(Q_2) = \text{initial } n(Q_2) + \text{change in } n(Q_2)$ }
Equilibrium $n(PQ) = \text{initial } n(PQ) - \text{change in } n(PQ)$ } ✓
- Divide equilibrium amounts of P and Q_2 and PQ by 2 dm^3 . ✓
- Correct K_c expression (formulae in square brackets). ✓
- Substitution of equilibrium concentrations into K_c expression. ✓
- Final answer: $10,89 / 10,889$ ✓

Nasienkriteria:

- Aanvanklike $n(P) = 4 \text{ mol}$ en $n(Q_2) = 2,4 \text{ mol}$ en $n(PQ) = 0$. ✓
- Verandering in $n(P) = \text{ewewigs } n(P) - \text{aanvanklike } n(P) = 2,8 \text{ mol}$. ✓
- GEBRUIK verhouding: $P : Q_2 : PQ = 2 : 1 : 2$ ✓
- Ewewig $n(Q_2) = \text{aanvanklike } n(Q_2) + \text{verandering in } n(Q_2)$ }
Ewewig $n(PQ) = \text{aanvanklike } n(PQ) - \text{verandering in } n(PQ)$ } ✓
- Deel ewewigshoeveelhede van P en Q_2 en PQ deur 2 dm^3 . ✓
- Korrekte K_c -uitdrukking (formules in vierkanthakies). ✓
- Vervanging van ewewigskonsentrasies in K_c -uitdrukking. ✓
- Finale antwoord: $10,89 / 10,889$ ✓

OPTION 3/OPSIE 3

	P	Q_2	PQ	
Initial quantity (mol) Aanvangshoeveelheid (mol)	4	2,4	0	✓(a)
Change (mol) Verandering (mol)	2,8 ✓(b)	1,4	2,8	✓(c)
Quantity at equilibrium (mol)/ Hoeveelheid by ewewig (mol)	1,2	1,0	2,8	✓(d)
Equilibrium concentration ($\text{mol} \cdot \text{dm}^{-3}$) Ewewigskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)	0,6	0,5	1,4	✓(e)

$$K_c = \frac{[PQ]^2}{[Q_2][P]^2} \quad \checkmark(f)$$

$$= \frac{1,4^2}{(0,5)(0,6)^2} \quad \checkmark(g)$$

$$= 10,89 \quad \checkmark(h)$$

No K_c expression, correct substitution/Geen K_c -uitdrukking, korrekte substitusie: Max./Maks. $\frac{7}{8}$

Wrong K_c expression/Verkeerde K_c -uitdrukking:
Max./Maks. $\frac{5}{8}$

CALCULATIONS USING CONCENTRATION**BEREKENINGE WAT KONSENTRASIE GEBRUIK****Marking criteria:**

- Initial $c(P)$ and $c(Q_2)$ and $c(PQ)$ from table. ✓
- Change in $c(P)$ = equilibrium $c(P)$ – initial $c(P)$. ✓
- USING** ratio: $P : Q_2 : PQ = 2 : 1 : 2$ ✓
- Equilibrium $c(Q_2)$ = initial $c(Q_2)$ + change in $c(Q_2)$ } ✓
Equilibrium $c(PQ)$ = initial $c(PQ)$ - change in $c(PQ)$ }
- Divide **initial** amounts of P and Q_2 and PQ by 2 dm^3 . ✓
- Correct K_c expression (formulae in square brackets). ✓
- Substitution of equilibrium concentrations into K_c expression. ✓
- Final answer: $10,89 / 10,889$ ✓

Nasienriglyne:

- Aanvanklike $c(P)$ en $c(Q_2)$ en $c(PQ)$ uit tabel. ✓
- Verandering in $c(P)$ = ewewigs $c(P)$ – aanvanklike $c(P)$. ✓
- GEBRUIK** verhouding: $P : Q_2 : PQ = 2 : 1 : 2$ ✓
- Ewewig $c(Q_2)$ = aanvanklike $c(Q_2)$ + verandering in $c(Q_2)$ } ✓
Ewewig $c(PQ)$ = aanvanklike $c(PQ)$ - verandering in $c(PQ)$ }
- Deel **aanvang**shoeveelhede van P en Q_2 en PQ deur 2 dm^3 . ✓
- Korrekte K_c -uitdrukking (formules in vierkanthakies). ✓
- Vervanging van ewewigskonsentrasies in K_c -uitdrukking. ✓
- Finale antwoord: $10,89 / 10,889$ ✓

OPTION 4/OPSIE 4

	P	Q_2	PQ	
Initial concentration ($\text{mol} \cdot \text{dm}^{-3}$)				
Aanvangskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)	0,4	0,4	1,6	✓ (a)
Change in concentration ($\text{mol} \cdot \text{dm}^{-3}$)				
Verandering in konsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)	0,2 ✓ (b)	0,1	0,2	✓ (c)
Equilibrium concentration ($\text{mol} \cdot \text{dm}^{-3}$)				
Ewewigskonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)	0,6	0,5	1,4	✓ (d)

$$K_c = \frac{[PQ]^2}{[Q_2][P]^2} \quad \checkmark \text{ (f)}$$

$$= \frac{1,4^2}{(0,5)(0,6)^2} \quad \checkmark \text{ (g)}$$

$$= 10,89 \quad \checkmark \text{ (h)}$$

No K_c expression, correct substitution/Geen K_c -uitdrukking, korrekte substitusie: Max./Maks. $\frac{7}{8}$

Wrong K_c expression/Verkeerde K_c -uitdrukking:
Max./Maks. $\frac{5}{8}$

(8)

6.2.4 Remains the same/Bly dieselfde ✓

Only temperature can change K_c /Temperature remains constant. ✓Slegs temperatuur kan K_c verander./Temperatuur bly konstant.

(2)

6.3

6.3.1 Increases/Toeneem ✓

(1)

6.3.2 Decreases/Afneem ✓

(1)

[18]

QUESTION 7/VRAAG 7

7.1

7.1.1 (It is a) proton/ H_3O^+ (ion)/ H^+ (ion) donor. ✓✓
(Dit is 'n) proton/ H_3O^+ -(ioon)/ H^+ -(ioon)skenker. (2)

7.1.2 HSO_4^- /hydrogen sulphate ion/waterstofsulfaatioon ✓

ANY ONE:

- It acts as base in reaction I and as acid in reaction II. ✓
- Acts as acid and base.

ENIGE EEN:

- Dit reageer as basis in reaksie I en as suur in reaksie II.
- Reageer as suur en basis. (2)

7.1.3 HSO_4^- /Reaction (solution) II/Reaksie (oplossing) II ✓



Smaller K_a value/weaker acid ✓

Lower ion concentration/Incompletely ionised. ✓

Kleiner K_a -waarde/swakker suur ✓

Laer ioonkonsentrasie/Onvolledig geïoniseer. ✓ (3)

7.2

7.2.1

<u>OPTION 1/OPSIE 1</u>	<u>OPTION 2/OPSIE 2</u>
$\text{pH} = -\log[\text{H}_3\text{O}^+]$ ✓ $1,02 = -\log[\text{H}_3\text{O}^+]$ ✓ $[\text{H}_3\text{O}^+] = 0,0955 \text{ mol}\cdot\text{dm}^{-3}$ ✓ Therefore/Dus $[\text{HCl}] = 0,0955 \text{ mol}\cdot\text{dm}^{-3}$ $(0,096/0,1 \text{ mol}\cdot\text{dm}^{-3})$	$\text{pH} = -\log[\text{H}_3\text{O}^+]$ } ✓ Any one/Enige $[\text{H}_3\text{O}^+] = 10^{-\text{pH}}$ $= 10^{-1,02}$ ✓ $= 0,0955 \text{ mol}\cdot\text{dm}^{-3}$ ✓ Therefore/Dus $[\text{HCl}] = 0,0955 \text{ mol}\cdot\text{dm}^{-3}$ $(0,096/0,1 \text{ mol}\cdot\text{dm}^{-3})$

(3)

7.2.2 **POSITIVE MARKING FROM 7.2.1/POSITIEWE NASIEN VAN VRAAG 7.2.1****Marking criteria:**

- Formula: $c = \frac{n}{V} / \frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$ ✓
 - Calculate $n(\text{Na}_2\text{CO}_3)$: $0,075 \times 0,025$ ✓
 - Calculate $n(\text{HCl})$: $0,0955 \times 0,05 / 0,096 \times 0,05$ ✓
 - Use ratios: $n(\text{HCl}) = 2n(\text{Na}_2\text{CO}_3)$ ✓
 - $n(\text{HCl})_{\text{excess}} = n(\text{HCl})_{\text{initial}} - n(\text{HCl})_{\text{used}} = 0,00475 - 0,0038$ ✓✓
 - Substitute $0,075 \text{ dm}^3$ in $c = \frac{n}{V}$ ✓
 - Final answer: $0,013 \text{ mol} \cdot \text{dm}^{-3}$ ✓ ($1,3 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3}$)
- Range:** 0,01 to 0,02 $\text{mol} \cdot \text{dm}^{-3}$

Nasienkriteria:

- Formule: $c = \frac{n}{V} / \frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$ ✓
 - Bereken $n(\text{Na}_2\text{CO}_3)$: $0,075 \times 0,025$ ✓
 - Bereken $n(\text{HCl})$: $0,0955 \times 0,05 / 0,096 \times 0,05$ ✓
 - Gebruik molverhouding: $n(\text{HCl}) = 2n(\text{Na}_2\text{CO}_3)$ ✓
 - $n(\text{HCl})_{\text{oormaat}} = n(\text{HCl})_{\text{aanvanklik}} - n(\text{HCl})_{\text{gebruik}} = 0,00475 - 0,0038$ ✓✓
 - Vervang $0,075 \text{ dm}^3$ in $c = \frac{n}{V}$ ✓
 - Finale antwoord: $0,013 \text{ mol} \cdot \text{dm}^{-3}$ ✓ ($1,3 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3}$)
- Gebied:** 0,01 tot 0,02 $\text{mol} \cdot \text{dm}^{-3}$

OPTION 1/OPSIE 1

$$\begin{aligned}
 n(\text{Na}_2\text{CO}_3) &= cV \checkmark \\
 &= 0,075 \times 0,025 \checkmark \\
 &= 0,001875 \text{ mol} && (1,875 \times 10^{-3} / 0,002 \text{ mol}) \\
 n(\text{HCl})_{\text{initial/aanvanklik}} &= cV \\
 &= 0,096 \times 0,05 \checkmark \\
 &= 0,00475 \text{ mol} && (4,75 \times 10^{-3} / 0,005 \text{ mol}) \\
 n(\text{HCl})_{\text{used/gebruik}} &= 2n(\text{Na}_2\text{CO}_3) \checkmark \\
 &= 2(0,001875) \checkmark \\
 &= 0,0038 \text{ mol} && (3,75 \times 10^{-3} / 0,004 \text{ mol}) \\
 n(\text{HCl})_{\text{excess/oormaat}} &= 0,00475 - 0,0038 \checkmark \checkmark \\
 &= 0,00095 \text{ mol} && (9,5 \times 10^{-4} / 1 \times 10^{-3} \text{ mol}) \\
 c(\text{HCl}) &= \frac{n}{V} \\
 &= \frac{0,00095}{0,075} \checkmark \\
 &= 0,013 \text{ mol} \cdot \text{dm}^{-3} \checkmark && (1,3 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3})
 \end{aligned}$$

OPTION 2/OPSIE 2

$$\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b} \checkmark$$

$$\frac{c_a(50)}{(0,075)(25)} = \frac{2}{1} \checkmark$$

$$c(\text{HCl})_{\text{rea}} = 0,075 \text{ mol} \cdot \text{dm}^{-3}$$

$$c(\text{HCl})_{\text{excess/oormaat}} = 0,0955 - 0,075 \checkmark \checkmark$$

$$= 0,0205 \text{ mol} \cdot \text{dm}^{-3}$$

$$c_1 V_1 = c_2 V_2$$

$$(0,0205)(50) = c_2(75) \checkmark$$

$$c_2 = 0,014 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

(8)
[18]**QUESTION 8/VRAAG 8**

- 8.1 Chemical (energy) to electrical (energy) $\checkmark$
Chemiese (energie) na elektriese (energie)

(1)

8.2

Marking criteria:

- Any formula: $c = \frac{m}{MV} / c = \frac{n}{V} / n = \frac{m}{M} \checkmark$
- Substitute $1 \text{ mol} \cdot \text{dm}^{-3} \checkmark$
- Substitute $170 \text{ g} \cdot \text{mol}^{-1}$ [or $108 + 14 + 3(16)$] and $0,15 \text{ dm}^3$ in correct formulae. $\checkmark$
- Final answer: $25,50 \text{ g} \checkmark$

Nasienkriteria:

- Enige formule: $c = \frac{m}{MV} / c = \frac{n}{V} / n = \frac{m}{M} \checkmark$
- Vervang $1 \text{ mol} \cdot \text{dm}^{-3} \checkmark$
- Vervang $170 \text{ g} \cdot \text{mol}^{-1}$ [of $108 + 14 + 3(16)$] en $0,15 \text{ dm}^3$ in korrekte formules. $\checkmark$
- Finale antwoord: $25,50 \text{ g} \checkmark$

OPTION 1/OPSIE 1

$$c = \frac{m}{MV} \checkmark$$

$$\checkmark 1 = \frac{m}{170 \times 0,15} \checkmark$$

$$m = 25,50 \text{ g} \checkmark$$

OPTION 2/OPSIE 2

$$n = cV \checkmark$$

$$= 1 \checkmark \times 0,15$$

$$= 0,15 \text{ mol} \checkmark$$

$$m = nM$$

$$= (0,15)(170)$$

$$= 25,50 \text{ g} \checkmark$$

(4)

8.3 **ANY ONE:**

- A substance that loses/donates electrons. ✓✓
- A substance that is oxidised.
- A substance whose oxidation number increases.

ENIGE EEN:

- 'n Stof wat elektrone verloor/skenk. ✓✓
- 'n Stof wat geoksideer word.
- 'n Stof wat waarvan die oksidasiegetal toeneem.

(2)

8.4

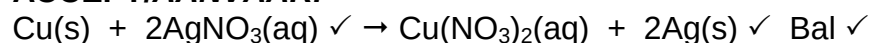
8.4.1 Copper/Cu/Koper ✓

(1)

8.4.2

Marking criteria/Nasienkriteria:

- Reactants ✓ Products ✓ Balancing ✓
Reaktanse Produkte Balansering
- Ignore double arrows./Ignoreer dubbelpyle.
- Ignore phases./Ignoreer fases.
- Marking rule 6.3.10./Nasienreël 6.3.10.

**ACCEPT/AANVAAR:****NOTE/LET WEL**

- IF electrons are not cancelled – minus 1 mark
- **INDIEN** elektrone nie gekanselleer is nie – minus 1 punt

(3)

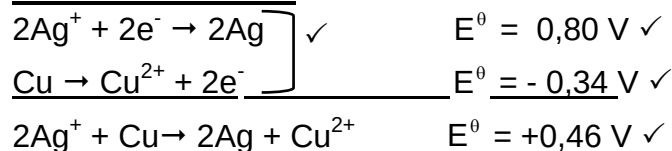
8.5

OPTION 1/OPSIE 1

$$\begin{aligned} E^\ominus_{\text{cell}} &= E^\ominus_{\text{reduction}} - E^\ominus_{\text{oxidation}} \checkmark \\ &= 0,80 \checkmark - (0,34) \checkmark \\ &= 0,46 \text{ V } \checkmark \end{aligned}$$

Notes/Aantekeninge

- Accept any other correct formula from the data sheet./Aanvaar enige ander korrekte formule vanaf gegewensblad.
- Any other formula using unconventional abbreviations, e.g. $E^\ominus_{\text{cell}} = E^\ominus_{\text{OA}} - E^\ominus_{\text{RA}}$ followed by correct substitutions:/Enige ander formule wat onkonvensionele afkortings gebruik bv. $E^\ominus_{\text{sel}} = E^\ominus_{\text{OM}} - E^\ominus_{\text{RM}}$ gevolg deur korrekte vervangings: $\frac{3}{4}$

OPTION 2/OPSIE 2

(4)

8.6 Decreases/Afneem ✓

(1)

[16]

QUESTION 9/VRAAG 9**9.1 ANY ONE: (2 or 0)**

- A substance whose (aqueous) solution contains ions. ✓✓
- Substance that dissolves in water to give a solution that conducts electricity.
- A substance that forms ions in water / when melted.
- A solution that conducts electricity through the movement of ions.

ENIGE EEN: (2 of 0)

- 'n Stof waarvan die oplossing ione bevat. ✓✓
- 'n Stof wat in water oplos om 'n oplossing te vorm wat elektrisiteit gelej.
- 'n Stof wat ione in water vorm/ wanneer dit gesmelt word.
- 'n Oplossing wat elektrisiteit gelej deur die beweging van ione. (2)

9.2 Anode ✓

Chromium is oxidised./Oxidation takes place (at the anode)./Chromium (it) loses electrons./Mass decreases./ $\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$ ✓
 Chroom word geoksideer./Oksidasie vind (by die anode) plaas./Chroom (dit) verloor elektrone./Massa neem af./ $\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$

NOTE/LET WEL:

If half-reaction is used, it must be correct./Indien halfreaksie gebruik word, moet dit korrek wees: $\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$ (2)

9.3 $\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Cr}(\text{s})$ ✓✓

Ignore phases./Ignoreer fases.

Marking guidelines/Nasienkriteria

- $\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$ $\frac{1}{2}$ $\text{Cr} \rightleftharpoons \text{Cr}^{3+} + 3\text{e}^-$ $\frac{0}{2}$
 - $\text{Cr} \leftarrow \text{Cr}^{3+} + 3\text{e}^-$ $\frac{2}{2}$ $\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$ $\frac{0}{2}$
 - Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron.
 - If charge (+) omitted on Cr^{3+} /Indien lading (+) weggelaat op Cr^{3+} : Max./Maks: $\frac{1}{2}$
- Example/Voorbeeld: $\text{Cr}^3 + 3\text{e}^- \rightarrow \text{Cr}$ ✓ (2)

9.4

Marking criteria:

- Substitute $52 \text{ g} \cdot \text{mol}^{-1}$ in $n = \frac{m}{M}$ /ratio ✓
 - Use mol ratio: $n(\text{electrons}): n(\text{Cr}) = 3 : 1$. ✓
 - Number of electrons = $n \times 6,02 \times 10^{23}$ /No of Cr atoms = $n \times 6,02 \times 10^{23}$ /ratio. ✓
 - Total charge = number of electrons $\times 1,6 \times 10^{-19}$ /ratio. ✓
 - Final answer: 11 113,85 C ✓
- Range:** 11 076,8 to 11 580 C

Nasienkriteria:

- Vervang $52 \text{ g} \cdot \text{mol}^{-1}$ in $n = \frac{m}{M}$ /verhouding ✓
 - Gebruik molverhouding: $n(\text{elektrone}) : n(\text{Cr}^{3+}) = 3 : 1$. ✓
 - Aantal elektrone = $n \times 6,02 \times 10^{23}$ /Aantal Cr-atome = $n \times 6,02 \times 10^{23}$ /verhouding. ✓
 - Totale lading = aantal elektrone $\times 1,6 \times 10^{-19}$ /verhouding. ✓
 - Finale antwoord: 11 113,85 C ✓
- Gebied:** 11 076,8 tot 11 580 C

OPTION 1/OPSIE 1

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{2}{52} \checkmark \\
 &= 0,038 \text{ mol} \quad (0,04 \text{ mol}) \\
 &\quad \swarrow \\
 n(e^-) &= 3n(\text{Cr}) \checkmark \\
 &= 3(0,038) \\
 &= 0,115 \text{ mol} \quad (0,12 \text{ mol}) \\
 &\quad \swarrow \\
 \text{Number } (e^-) &= 0,115 \times 6,02 \times 10^{23} \checkmark \\
 &= 6,946 \times 10^{22} \\
 &\quad \swarrow \\
 Q &= 6,95 \times 10^{22} \times 1,6 \times 10^{-19} \checkmark \\
 &= 11\,113,85 \text{ C} \checkmark
 \end{aligned}$$

OPTION 2/OPSIE 2

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{2}{52} \checkmark \\
 &= 0,038 \text{ mol} \quad (0,04 \text{ mol}) \\
 &\quad \swarrow \\
 \text{Number Cr atoms} &= 0,038 \times 6,02 \times 10^{23} \checkmark \\
 &= 2,315 \times 10^{22} \\
 &\quad \swarrow \\
 \text{Number } (e^-) &= 3N(\text{Cr}) \checkmark \\
 &= 3(2,315 \times 10^{22}) \\
 &= 6,946 \times 10^{22} \\
 &\quad \swarrow \\
 Q &= 6,95 \times 10^{22} \times 1,6 \times 10^{-19} \checkmark \\
 &= 11\,113,85 \text{ C} \checkmark
 \end{aligned}$$

OPTION 3/OPSIE 3

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{2}{52} \checkmark \\
 &= 0,038 \text{ mol} \\
 &\quad \swarrow \\
 n(e^-) &= 3n(\text{Cr}) \checkmark \\
 &= 3(0,038) \\
 &= 0,115 \text{ mol} \\
 &\quad \swarrow \\
 1 \text{ mol} &\quad \dots \quad 96\,500 \text{ C} \checkmark \\
 0,115 \text{ mol} &\quad \quad \quad 11\,134,62 \text{ C} \checkmark \checkmark
 \end{aligned}$$

(5)
[11]**TOTAL/TOTAAL:****150**